# 📦 Enterprise Inventory Management & Analytics System

A production-grade, modular Inventory Management Application built with **Python**, **MySQL**, **Pandas**, and **Matplotlib**. Features parameterized SQL queries, secure environment configurations, transactional stock updates, automated CSV reporting, and visual data analytics.

## 📁 Repository Structure

```
.
├── database.py        # Database connections and schema initialization
├── operations.py      # Product CRUD and transaction business logic
├── analytics.py       # Pandas queries, CSV exports, and Matplotlib
visualizations
├── main.py            # CLI interactive loop
├── .env.example       # Sample environment variable template
├── requirements.txt   # Python package dependencies
└── README.md          # Project documentation
```

## ✨ Features

- **Modular Architecture:** Code separation across data storage, operations, and analytical rendering layers.
- **Security & Parametrically Safe SQL:** Relies on .env files for DB credentials and utilizes parameterized SQL queries to prevent SQL injection vulnerabilities.
- **Transactional Integrity & Validation:** Automatically validates input types, handles out-of-stock edge cases, and logs transaction timestamps.
- **Data Visualizations:** Built-in charts displaying item stock distributions and daily unit sales volume trends.
- **Enterprise Reporting:** Exports sales and purchase histories directly to structured .csv files.

## 🛠️ Installation & Setup

1. **Clone the Repository:**
   ```
   git clone
   https://github.com/your-username/inventory-management-system.git
   cd inventory-management-system
   ```

2. **Setup Virtual Environment:**
   ```
   python -m venv venv
   source venv/bin/activate  # On Windows: venv\Scripts\activate
   ```

3. **Install Dependencies:**
   `pip install -r requirements.txt`

4. **Configure Environment Variables:**
   - Duplicate .env.example to create a .env file:
     `cp .env.example .env`

   - Fill in your local MySQL connection details inside .env.
5. **Run the Application:**
   `python main.py`